Which Tool? · check your sort

Lesson 3 · LT 4.4 · Card A needs two tools, in order. Card F has no tool that works.

A	A drop of blood is left on broken glass at a shed. Police want to know whose blood it is.	**PCR, then gel**	PCR copies the tiny sample; the gel compares the bands with each suspect's.
B	A bone from the ice holds only a trace of mammoth DNA. Scientists need enough to study.	**PCR**	PCR copies a tiny amount of DNA into millions of copies.
C	A company wants big tanks of bacteria that make human insulin.	**Plasmid + ligase**	Cut the insulin gene and a plasmid with one enzyme, seal with ligase, put it in bacteria.
D	Doctors want to break the switch gene that turns off fetal hemoglobin in a patient's blood stem cells.	**CRISPR**	The guide RNA finds the switch gene; Cas9 cuts it there.
E	A scientist wants to check whether a plasmid got bigger after a gene was pasted in.	**Gel**	A bigger plasmid runs a shorter distance than the plain plasmid.
F	Identical twins both visited the scene. Which twin left the hair?	**No tool here**	Identical twins have the same DNA, so their bands match each other.
G	Farmers want corn plants that make their own caterpillar poison.	**Plasmid + ligase**	A bacterial poison gene is pasted into the plant's DNA: a GMO.
H	A clinic has a nose swab and needs to know if a virus is in it.	**PCR**	PCR copies the virus's genes if they are there; no virus, no copies.

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